



CROSS EXAM / HACKGT 13

HEARSAY

Is this voice real?

CHALLENGE

NSA HEARSAY

CREATED BY

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DELIVERABLE

Audio authentication



AGENDA

01

OVERVIEW

02

APPROACH & SCOPE

03

RESULTS

04

DETECTORS

05

WHAT WE LEARNED

06

NEXT STEPS

PROJECT OVERVIEW



PROBLEM

A convincing synthetic voice can sound real. Calling real speech synthetic is the expensive error in this challenge.

SOLUTION

One decode path, ten forensic detectors, a fixed router, and an explained probability for every file.

MEASURED RESULTS

0.0065 holdout · 0.228 In-the-Wild normalized minDCF.



PROJECT OVERVIEW

Give HEARSAY an audio file. It returns a score, the evidence behind it, and a record



of how the decision was made.

THE OUTPUT

A submission TSV and one explanation JSON per file.

THE PRINCIPLE

Every detector runs; the router decides which findings may affect the score.

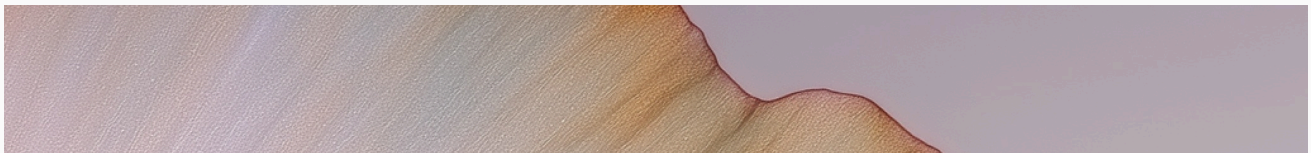
SCOPE

CORE SYSTEM

- Decode audio once to a shared format
- Run ten detectors behind one contract
- Fuse M1b, handcrafted, and M5 by rank
- Let Spectra only lower a likely false alarm
- Write a score and evidence report

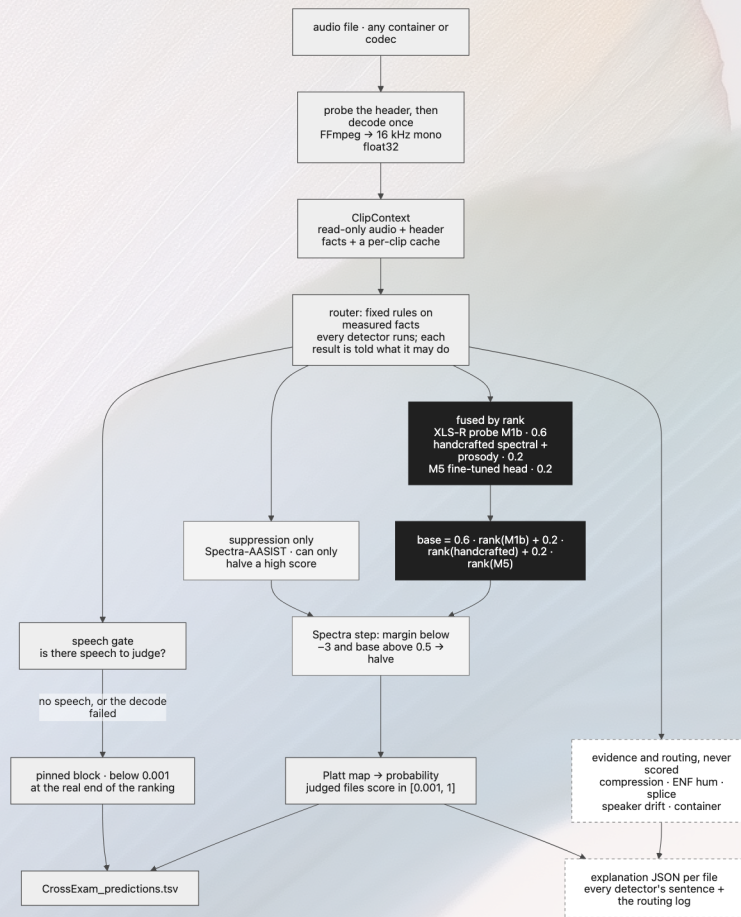
OUT OF SCOPE

- No hardware signal or device attestation
- No hosted model on the scoring path
- No retraining during inference





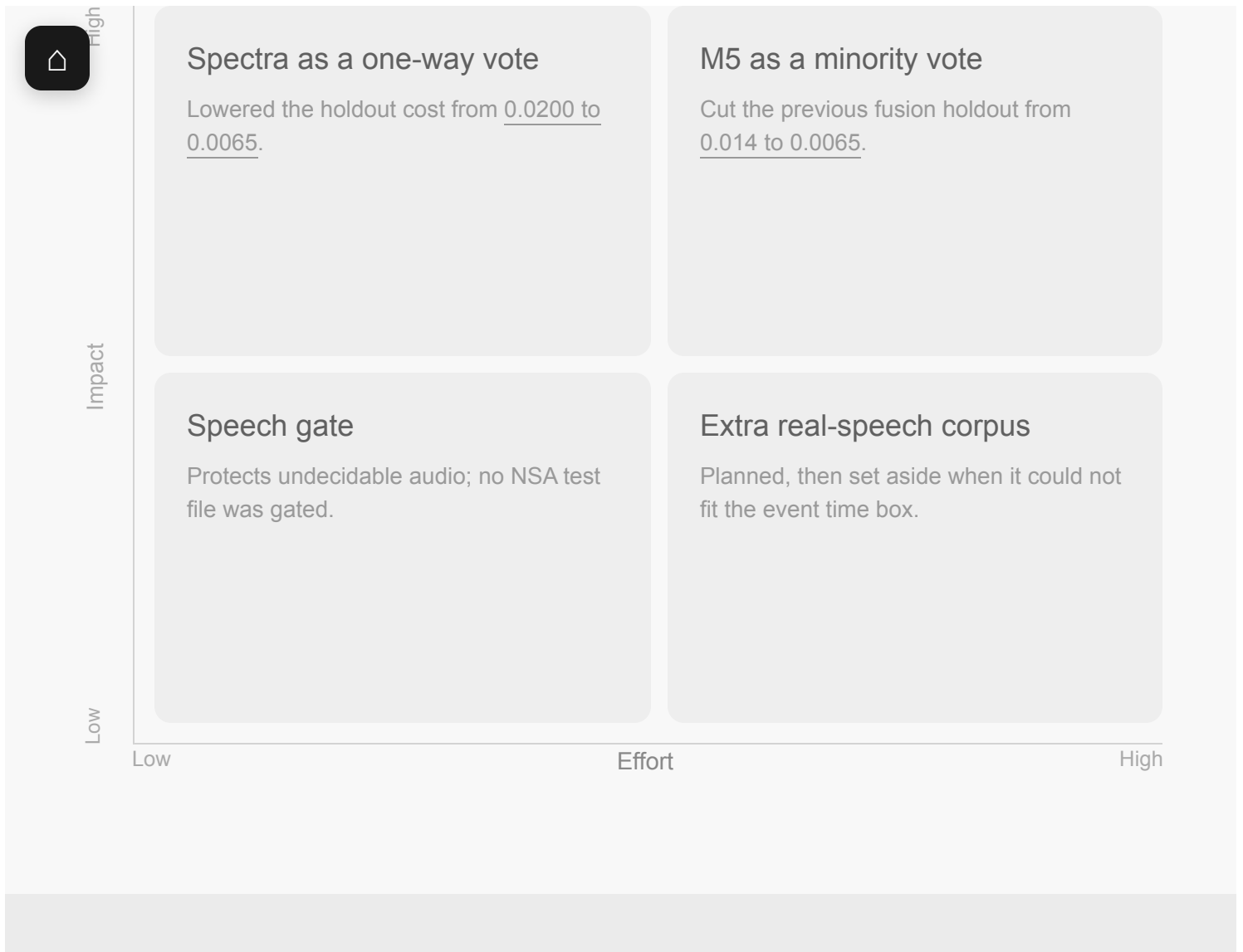
THE SHIPPED PIPELINE



[Read the full method ↗](#)

IMPACT VS EFFORT

Each technique had to earn its place in the shipped rule.



TECHNICAL APPROACH

MODEL STACK

- XLS-R layer-7 probe (M1b)
- Handcrafted spectral and prosody model
- Fine-tuned XLS-R head (M5)
- Spectra-AASIST as a suppressor

ARCHITECTURE PRINCIPLES

1. Decode once for consistent inputs
2. Run every detector and route by role
3. Fuse ranks, then calibrate
4. Keep inference local and explainable



MILESTONES

PHASE 01

FOUNDATIONS & DETECTORS

- Common audio loader and detector contract
- Fold file and the XLS-R probe
- Engineered forensic techniques

Comparable detector scores and evidence

PHASE 02

ROUTING & FUSION

- Pre-declared selection rule
- Rank blend and Spectra suppression
- Holdout and In-the-Wild validation

An auditable shipped

PHASE 03

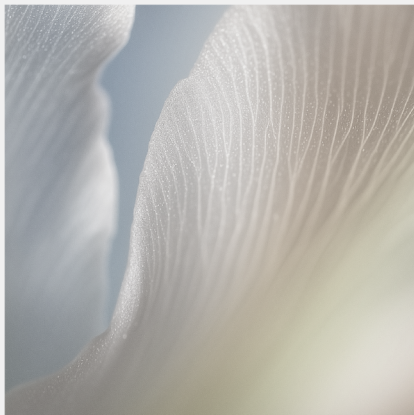
SUBMISSION & REVIEW

- Run the full test set
- Write the TSV and explanation JSON
- Check parity and document limits

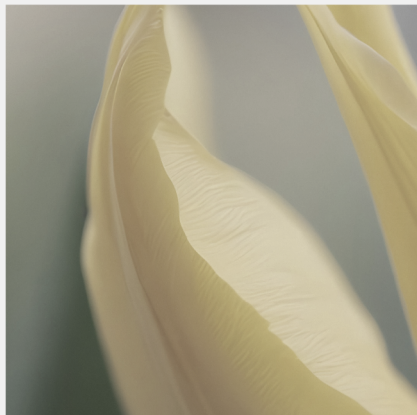
Reproducible output and documentation



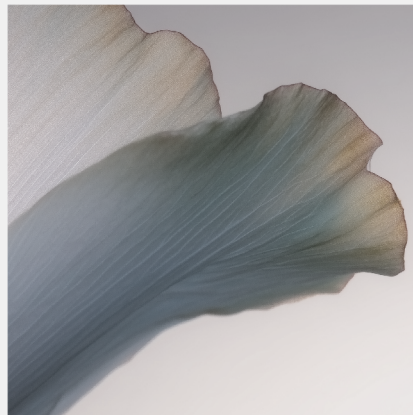
DETECTORS



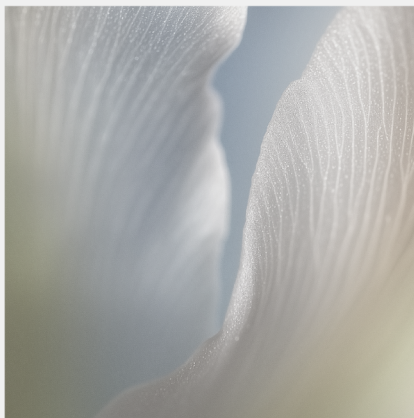
M1b
Primary probe



Handcrafted
Spectral evidence



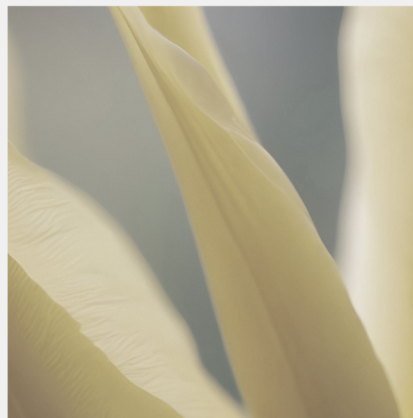
M5
Fine-tuned head



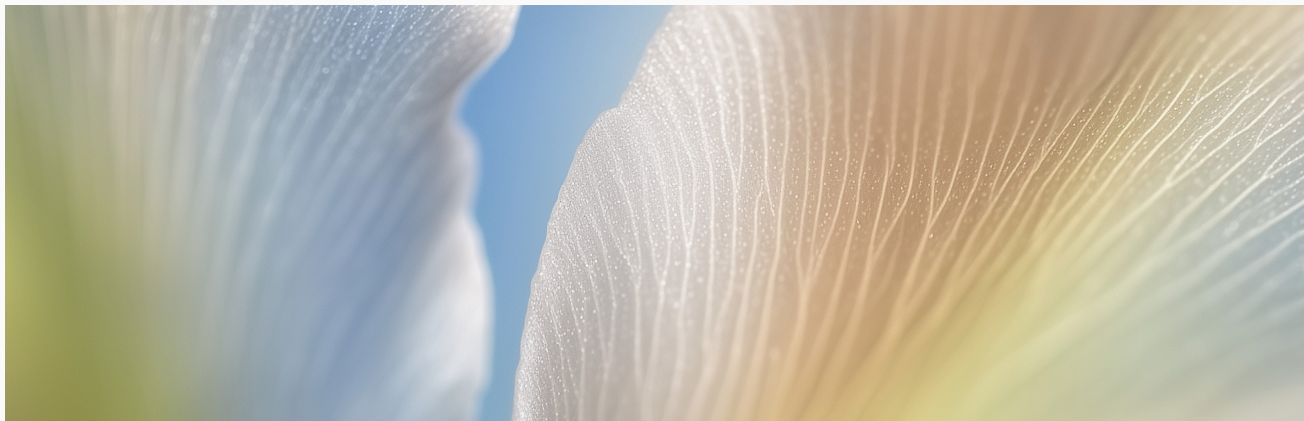
Spectra
One-way suppressor



Speech gate
Default-answer policy



Forensics
Evidence-only signals





WAYS OF WORKING

DECISION RULE

We wrote the fusion selection rule before viewing results and applied it once. Rejected ideas stay in the record.

AUDIT TRAIL

Each file keeps detector reasons and a routing log. Sources, tests, and known limits are linked throughout the docs.

THE LESSONS / 01-04

WHAT WE LEARNED

01 / 02

01

Our test data lied to us, five times.

Real and fake clips differed in volume, silence at the start, length, file format, and a filter cutoff at 7.2 kHz. A model could “win” on any of those without hearing the voice. We found each one and removed it. Next time we check for this on day one.

02

Good scores at home don't mean good scores in the wild.

Our best models scored around 0.1 on our own held-out data and around 0.4 on real-world recordings. The fusion rule that looked best at



home was the worst in the wild. We only avoided shipping it because we wrote down the selection rule before seeing the numbers.

03

Fix the data before adding more of it.

Adding 40 real speakers made things worse until we matched our training audio to the test set's filter. Then the same speakers made things better.

04

One false alarm costs 9 misses. Design for that.

This one number shaped everything: a detector that can only lower a score, a model tuned for fewer false alarms, and unreadable files pinned to the bottom of the list.

[Explore the evidence ↗](#)

THE LESSONS / 05-08

WHAT WE LEARNED

02 / 02

05

Half our detectors don't vote, and that's fine.

Five of ten detectors turned out to detect the recording setup, not the voice. Mains hum means “recorded at home”. Compression means “went through a pipeline”. Those detectors now explain the answer instead of setting it.

06

A weak model can still help.

Our fine-tuned model lost to the simple one on every test. But it makes different mistakes, so giving it a small vote cut errors in half.



07

If you can't check a model, limit what it can do.

Spectra-AASIST scored perfectly on our data, but nobody knows what it was trained on. So it can only say “this is real”, never “this is fake”.

08

Test for weaknesses before you freeze.

We found that background noise breaks the shipped model after it was too late to fix.

[Explore the evidence ↗](#)

NEXT STEPS

READ THE WORK

1. [Walk through the decision path](#)
2. [Inspect validation and limits](#)
3. [Explore all ten detectors](#)
4. [Reproduce the submitted output](#)

NSA READOUT

0.0733 minDCF

NSA returned 3.534% EER for the exact file we ship. The result fell in our pre-declared 0.00–0.20 band, so we kept the rule unchanged.

[Read the validation record ↗](#)



Continue into the complete
HEARSAY documentation.

Created by Nathan, Hrushi & Tharakesh.

[Open the full record ↗](#)

[Browse the library ↗](#)

THANK YOU.



FULL DOCUMENTATION AND SOURCES

ON THIS PAGE

- 01 The numbers
- 02 The cost function we are judged on
- 03 How it decides, in one picture
- 04 Results at a glance
- 05 Where to go next

THE NUMBERS

All figures are normalized minDCF (lower is better). The rule was chosen by a pre-declared selection rule, applied once; the out-of-fold figure is the training-side

readout, the holdout and In-the-Wild figures were each read once after the choice.



OUTER HOLDOUT

0.0065[§]

two unseen generators and 26 unseen real-speech groups

IN-THE-WILD, BRIEF COST

0.228[§]

web-sourced speech we never trained on; 0.239[§] under the sponsor code's cost

INNER OUT-OF-FOLD

0.135[§]

pooled over the five inner folds

NSA TEST SET, DRAFT REVIEW

0.0733[§]

returned minDCF; EER 3.534%[§] on the submitted file

THE COST FUNCTION WE ARE JUDGED ON

$$\text{minDCF} = 9.33 \times P_{\text{FA}} + 1 \times P_{\text{miss}}$$

One real file called synthetic costs as much as 9.33 missed fakes. That single ratio shaped every decision in this system.

P_{FA}

false-alarm rate: the share of real files scored above the threshold



miss

miss rate: the share of synthetic files scored below it

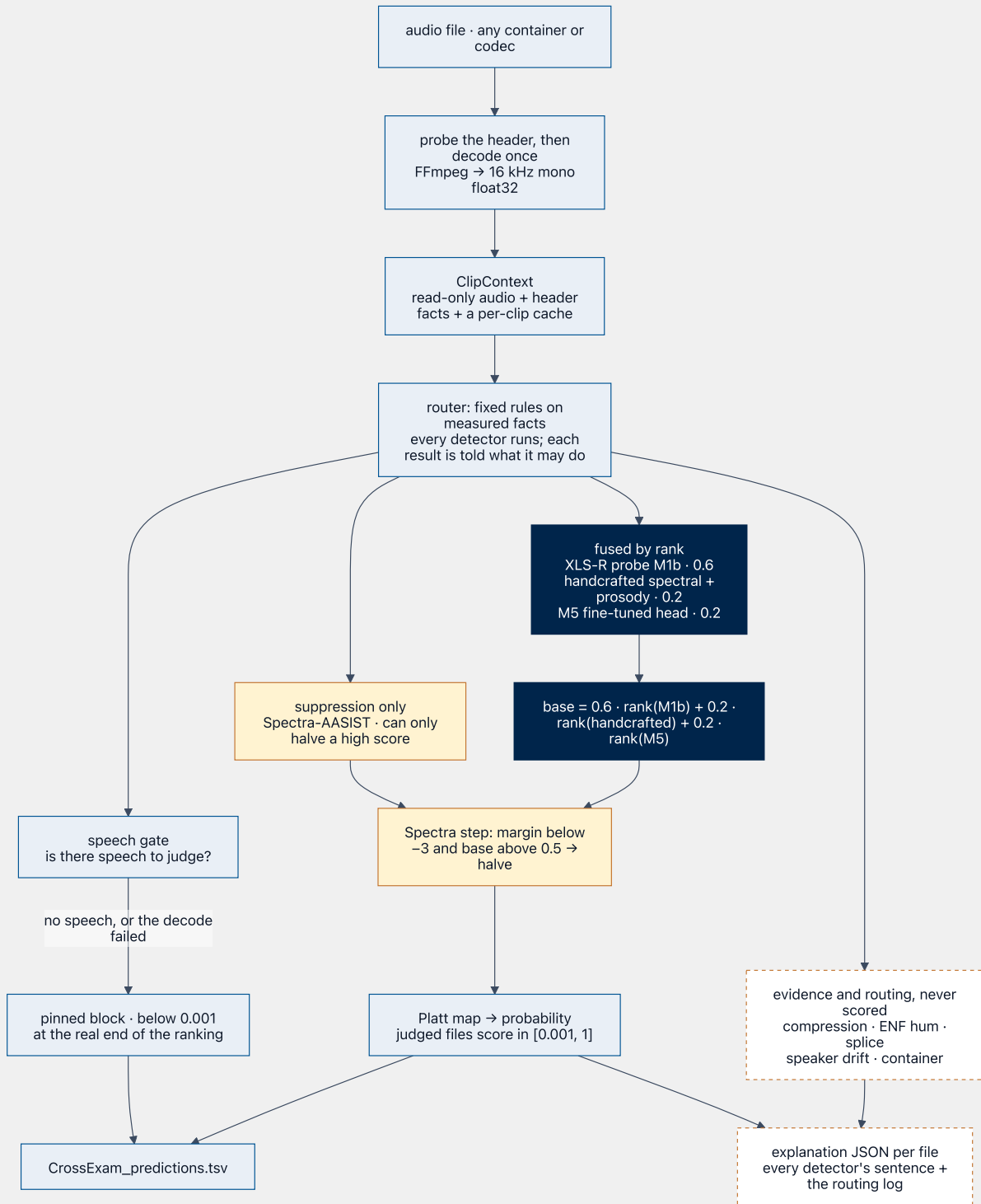
min

the best threshold is chosen by the scorer, so only the ranking of scores matters

Built from the sponsor's weights $C_{FA} = 4$, $C_{miss} = 1$ and a prior of 0.3 synthetic (about 0.7 of files are real):
cost = $4 \cdot P_{FA} \cdot 0.7 + 1 \cdot P_{miss} \cdot 0.3 = 2.8 \cdot P_{FA} + 0.3 \cdot P_{miss}$, divided by the best constant decision, 0.3, so
that 1.0 means "no better than always saying real" and 0.0 is perfect. ^{\$}

HOW IT DECIDES, IN ONE PICTURE

One decode path, ten detectors behind one contract, a router that tells each result what it may do, and a fusion rule that reads only ranks: $0.6 \times \text{rank}(\text{M1b}) + 0.2 \times \text{rank}(\text{handcrafted}) + 0.2 \times \text{rank}(\text{M5})$, then Spectra-AASIST allowed only to pull a file toward real, then a Platt map, with undecidable files in the pinned block at the real end. The full walk-through, with a real test file, is on How it decides.



The shipped pipeline: decode once, route, ten detectors, three fused by rank, Spectra as a one-way vote, Platt map, pinned block, TSV and explanation JSON. [Open full size](#)

► Mermaid source



RESULTS AT A GLANCE

	INNER OOF	HOLDOUT
Shipped rule (<u>A3 w 0.2 + E</u> : M1b, handcrafted and M5 by rank, Spectra suppression)	0.135 [§]	0.0065 [§]
Previous rule (E on α 0.2 [§] , no M5), the fallback	0.140 [§]	0.014 [§]
<u>M1b</u> alone, the best single detector we trained	0.301 [§]	0.072 [§]
<u>Spectra-AASIST</u> off the shelf, a <u>suppressor</u> only	not fused: its training data is undisclosed, so no row of ours is provably out-of-sample for it	

The test set has 1,671[§] files. The shipped rule scores 27.4%[§] of them above 0.5[§] against the roughly 30%[§] synthetic the brief states (the orchestration ablation, a different readout, reports 27.5%[§]); the speech gate abstains on 0 of 1,671[§]. Turning the router off moves the holdout from 0.0065[§] to 0.0200[§] and In-the-Wild from 0.229[§] to 0.274[§].

WHERE TO GO NEXT



How it decides

The pipeline, the router and the fusion arithmetic, on a real test file with its evidence sentences.

The metric and the default answer

What minDCF is, why a false alarm costs so much, why only ranking matters, and where the undecidable files go.

What worked and what did not

The honest log: every shortcut found, every idea rejected, with its number and its mechanism.

The ten detectors

One card each, mapped to the sponsor's rubric, with its role, its numbers and one real evidence sentence.

Validation

Folds by generator and speaker, the pre-declared sweeps, and the decoding table for the draft review.

Systems and subsystems

One spec page per part of the system, every parameter cited to the line that sets it.

The library

Every document in the repository, rendered: reports, run specs, audits, handoffs, consult records.

Reproduce it

The commands, verbatim, and where every artifact lives.

Generated from docs/site/source/pages/index.frag d976546b6cd4, README.md 937c075556d3. The
kdown is the record. [About this site](#).



HOW IT DECIDES

Every file takes the same road: decoded once, handed to ten detectors, routed by fixed rules, fused by rank, checked by one model that may only lower a score, mapped to a probability, and written to the TSV with a report that says why. This page walks that road with one real test file.

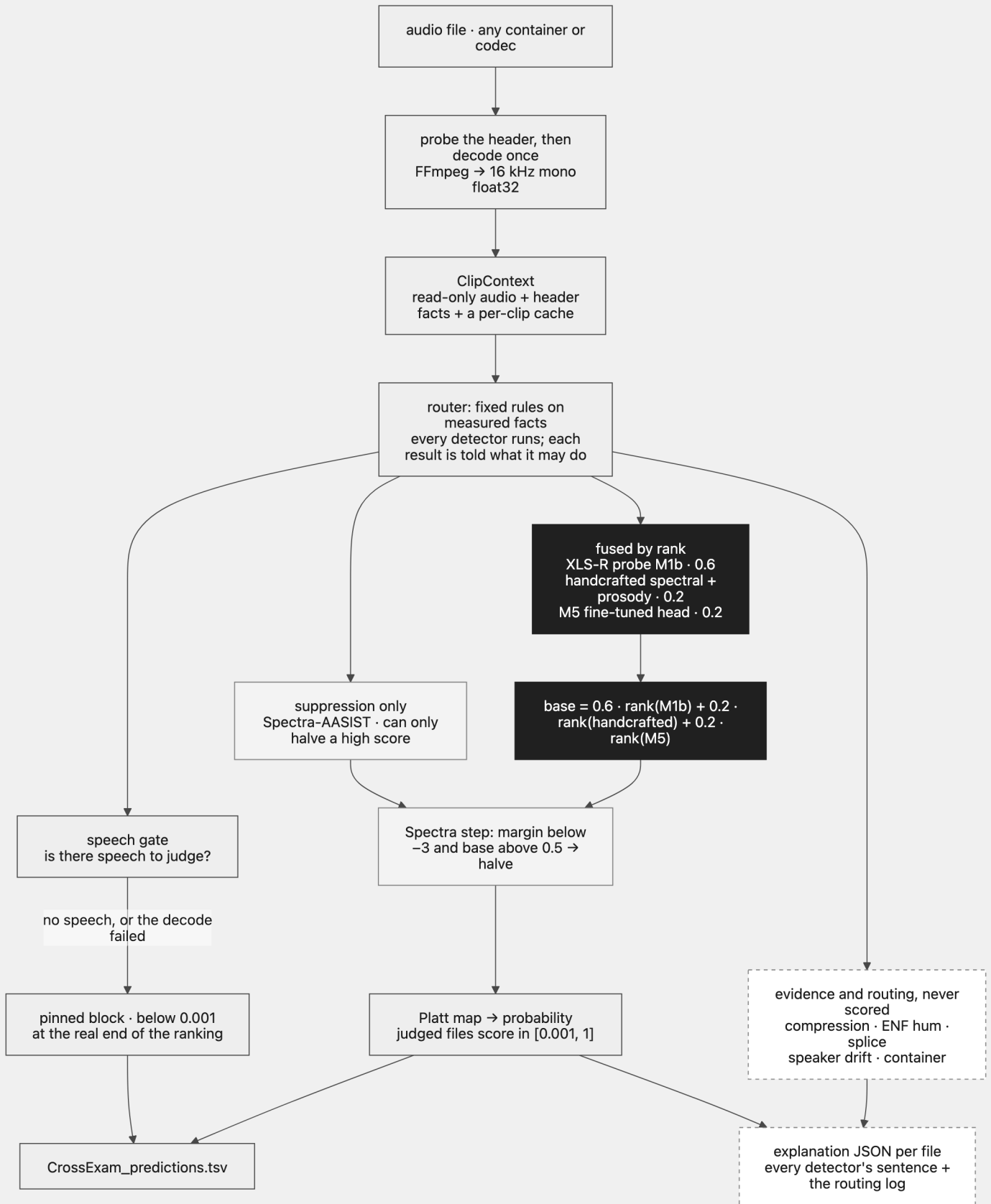
ON THIS PAGE

- 01 The six steps
 - 02 The router: which fact decides what
 - 03 One file, end to end
-

THE SIX STEPS

1. **Decode once.** Every file goes through one FFmpeg path to 16 kHz mono float32, and the header facts are read before decoding. Nothing downstream ever sees the original container, sample rate or filename.
2. **Run every detector.** Each detector gets the same read-only clip and returns a score in [0, 1] (higher = more synthetic), a one-sentence reason, named features and a status. A detector that crashes becomes an error at 0.5[§] and is treated as missing, never as evidence, and never costs a row.
3. **Route.** The router is a set of fixed rules over measured file properties. It does not choose which detectors run (the whole test set is about 95 minutes of audio, so everything runs); it decides what each result is *allowed to do*: be fused, only lower the score, or stand as evidence. Every decision is written to the file's routing log.

4. **Fuse by rank, and let Spectra only pull scores down.** The shipped rule is $\frac{0.6 \times \text{rank}(\text{M1b}) + 0.2 \times \text{rank}(\text{handcrafted}) + 0.2 \times \text{rank}(\text{M5})}{3}$, each rank taken against that detector's own out-of-fold training scores. Spectra-AASIST can halve a high score when it is confident the voice is real. It can never raise one.
5. **Abstain at the bottom.** A file with no speech to judge, or one that fails to decode, is pinned below every scored file. With a false alarm costing 9.33[§] misses, "we don't know" belongs at the real end of the ranking.
6. **Write the TSV and one JSON per file.** Header `filename<TAB>cm-score`, rows in the template's order, scores in [0.001[§], 1] for every file we could judge, and an explanation JSON beside each.

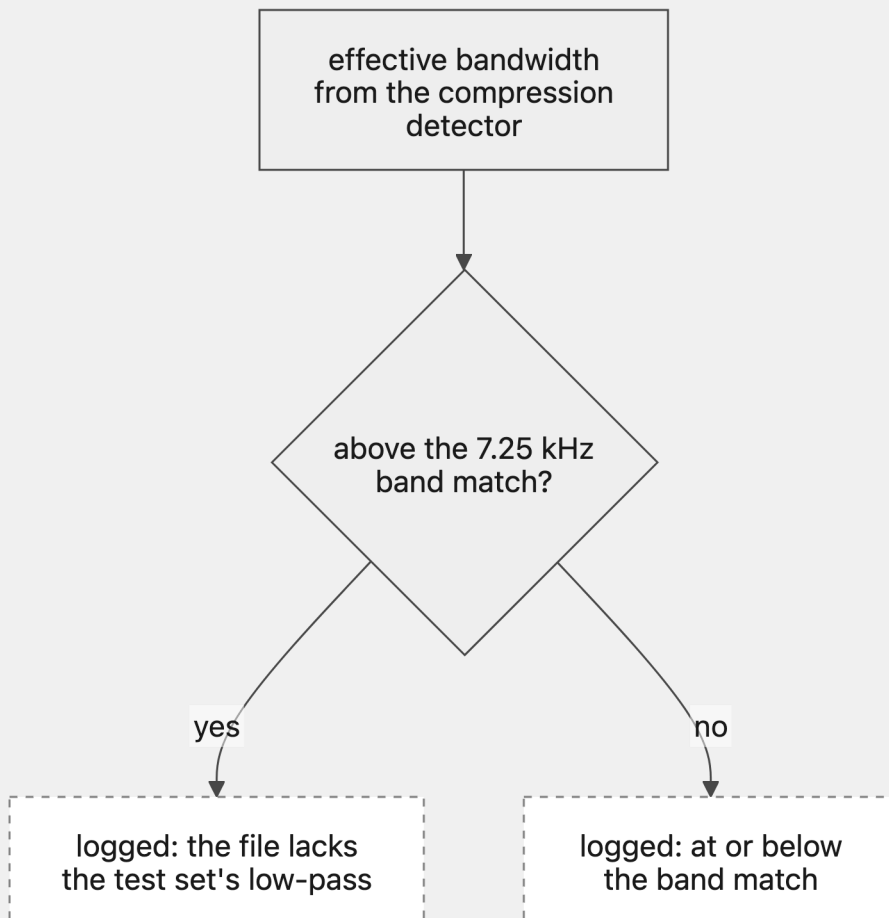
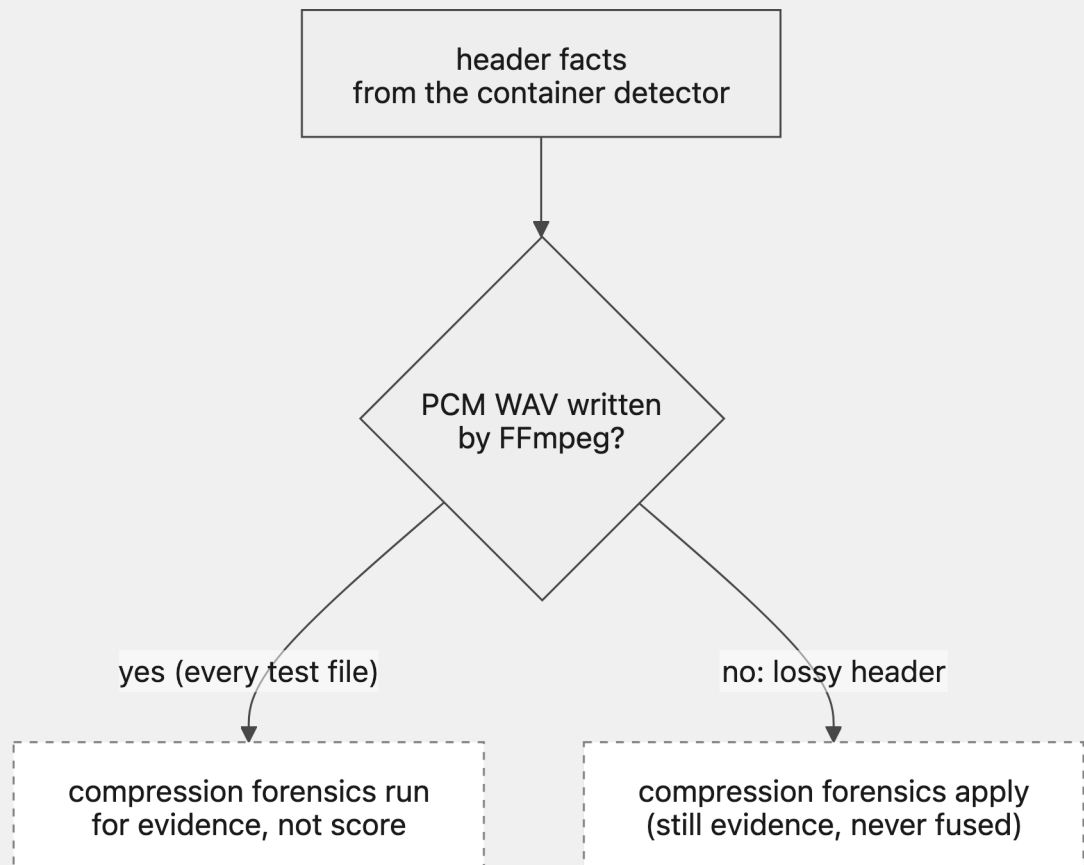


The shipped pipeline from audio file to TSV and explanation JSON. [Open full size](#)

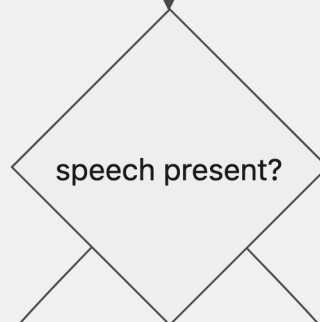
► Mermaid source

THE ROUTER: WHICH FACT DECIDES WHAT

Orchestration here is not a learned model and not a language model. It is a short list of rules over facts the detectors measure, and the point of each rule is written next to it. The per-file routing log records every decision, so a reader can check any file after the fact.



speech gate
five cues, thresholds beyond
the test set's extremes



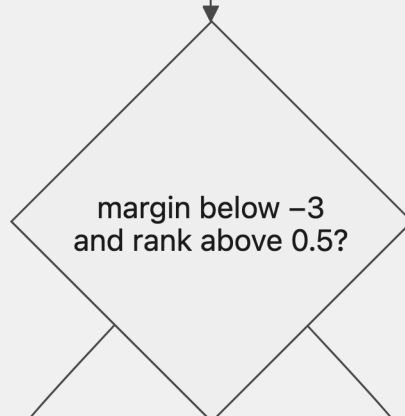
yes

fusion runs

no

skip fusion →
pinned block, real end

Spectra margin
+ fused rank



yes

rank halved:
a likely false alarm

no

unchanged:
Spectra never promotes

hum · seams · drift

written to the routing log
and the evidence list only

a fused detector errored?

its column imputed at the
training mean; the log names
it

The router's rules: container facts, effective bandwidth, the speech gate, the Spectra step, evidence-only detectors, and imputation of a missing fused column. [Open full size](#)

► Mermaid source

Router on versus off, measured

The orchestration ablation re-fuses the exported detector scores with each rule switched on and off. "Router off" is the plain rank blend with no Spectra suppression, no gate and no pinned block; "fuse everything equally" is a four-way equal rank mean of M1b, the handcrafted model, M5 and Spectra.

SPLIT	ROUTER ON (SHIPPED)	NO SPECTRA SUPPRESSI ON	ROUTE R OFF	FUSE EVERYTHING EQUALLY	FILES SPECTRA SUPPRESSION TOUCHED	FILES THE GATE TOUCHED
Holdout, 3,858 [§] rows	0.0065 [§]	0.0200 [§]	0.02 00 [§]	0.0045 [§]	83 (all real)	31
In-the-Wild, brief cost	0.229 [§]	0.274 [§]	0.27 4 [§]	0.214 [§]	24 (all real)	9

SPLIT	ROUTER ON (SHIPPED)	NO SPECTRA SUPPRESSI ON	ROUTE R OFF	FUSE EVERYTHING EQUALLY	FILES SPECTRA SUPPRESSION TOUCHED	FILES THE GATE TOUCHED
In-the-Wild, sponsor- code cost	0.2415 [§]	0.2535 [§]	0.25 05 [§]	0.1955 [§]		

NSA test, share above 0.5 [§]	27.5% [§]	30.3% [§]	30.3 % [§]	28.8% [§]	47	0
--	--------------------	--------------------	------------------------	--------------------	----	---

- **Spectra suppression is the rule that changes decisions.** It cuts the holdout cost to a third and takes 0.045[§] off In-the-Wild under the brief's cost, and every file it touched where a label exists was real.
- **The gate is a measured null on the test set.** It touched holdout and In-the-Wild rows and 0 test files; its thresholds were set beyond the test set's extremes on purpose, so it only catches silence, tones, static and noise.
- **The evidence-only detectors never move a score, by design.**
- **Our rule does not win every readout.** With M5 in the blend, fusing everything equally beats the shipped rule on every labeled readout. It does so by giving Spectra a full vote, which we ruled out before seeing these numbers because Spectra's training data is undisclosed. The choice rests on that argument, and the numbers are reported as measured.

ONE FILE, END TO END

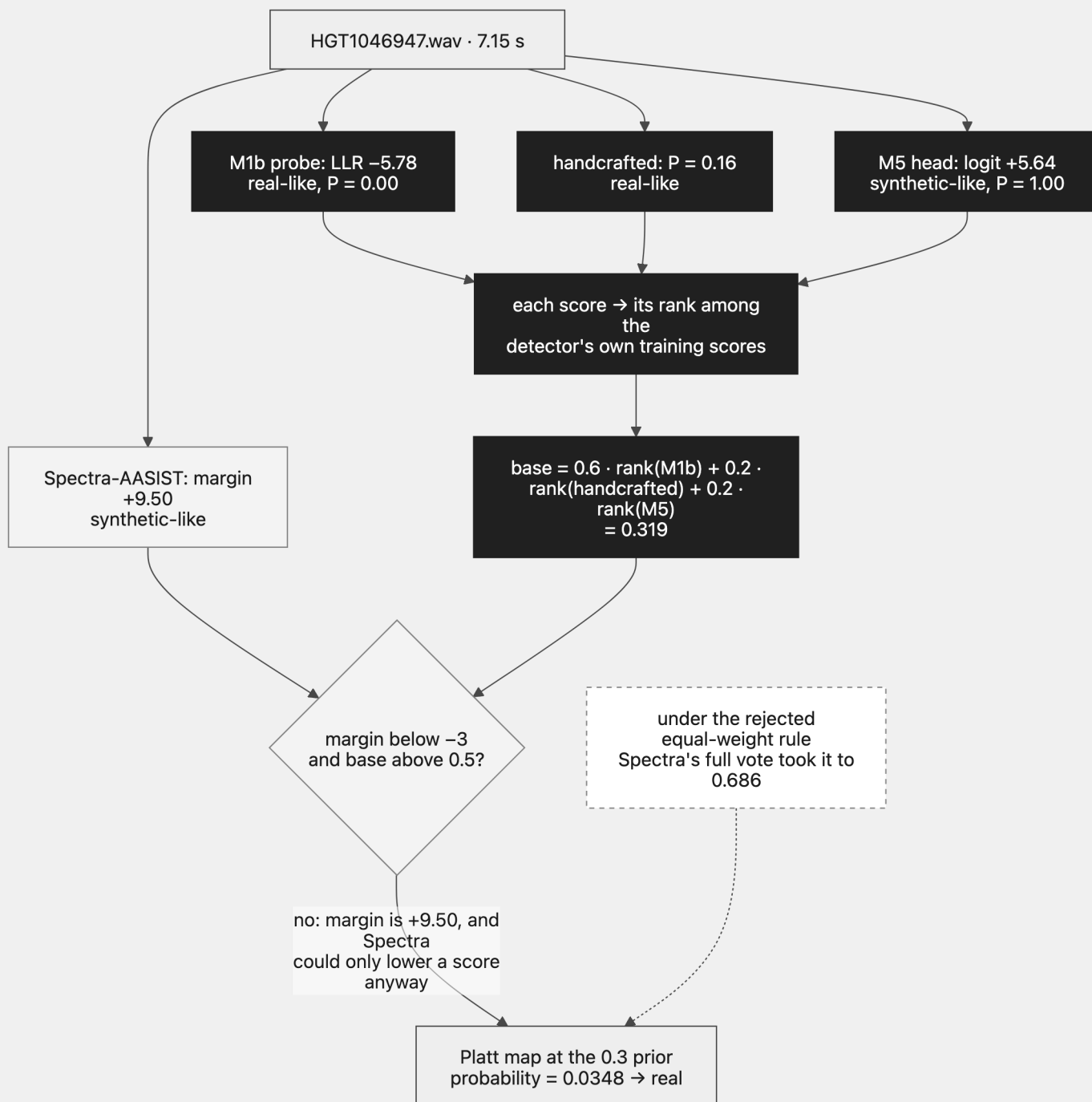
Test file `HGT1046947.wav` is one where the deep detectors disagree. What follows is the shipped pipeline's own output, run live from audio; its score matches the submitted TSV to seven millionths. [Source: the README's trace.](#)

fused	m1b_v3	XLS-R layer-7 probe: calibrated log-likelihood ratio -5.78 (real)
fused	m5_xlsr_ft	XLS-R fine-tuned head (M5, 12 layers, attentive pooling): logit
fused	handcrafted	real-like (P=0.16): LFCC 16 frame-to-frame change 3.0 SD below r
fused	spectra_aassist	spoof-minus-bonafide margin +9.50 (synthetic-like, P=1.00)
evidence	speaker_drift	voice drifts across the clip: minimum window-to-window speaker s
evidence	compression	compression-trace features synthetic-like (P=0.91): ...
routing	container	wav/pcm_s16le 16000 Hz mono, [encoder=Lavf58.29.100]: no class e
gate	speech_gate	speech present: voiced 28% of frames, pitch spread 0.34, loudnes

routing_log:

container: PCM WAV, not lossy, FFmpeg-written; compression forensics run for evidence,
 compression: effective bandwidth 7500 Hz, above the 7.25 kHz band match
 enf: no mains hum; no environment evidence either way
 splice: no editing seams
 speaker_drift: voice drifts (min window similarity -0.10); evidence only, never fused
 speech_gate: is_speech=true (voiced 28% of frames); default-answer policy not applied
 fusion: e_on_a (A3_w0.2_E): $0.6 \times \text{rank}(\text{m1b_v3}) + 0.2 \times \text{rank}(\text{handcrafted_v5}) + 0.2 \times \text{rank}(\text{evidence_v3})$
 M3 margin +9.50, no suppression (M3 never promotes)

probability_synthetic = 0.0348 -> real



The fusion rule as arithmetic on HGT1046947.wav: three ranks blended, the Spectra step not applied, the Platt map, and what the rejected equal-weight rule would have said. [Open full size](#)

► Mermaid source

Spectra-AASIST and M5 both say this voice is fake; the probe and the handcrafted model say real. Under the equal-weight rule we first tried, Spectra's vote carried the file to 0.686[§], above the midpoint. Under the shipped rule, M5's 20%[§] share lifts it from 0.0027[§] (the previous rule, without M5) to 0.0348[§],

still firmly real. Spectra gets no say in that direction: we do not let a pretrained model with undisclosed training data push a file toward "synthetic", because being wrong that way is the expensive error. The drift and compression lines stay in the report as evidence and never touch the score.

Seven more files

The worked-examples report walks eight real test files through the pipeline with their full routing logs and every evidence sentence: a confident real, a confident fake, the file where Spectra's suppression fired, a fake with a stable mains hum, a real file with editing seams, a 21%[§]-voiced file near the gate, and a genuinely uncertain one. They were written on the previous rule and re-run under the shipped rule; no file changes side of 0.5[§]. Source: the addendum.

FILE	SITUATION	P UNDER THE PREVIOUS RULE	P UNDER THE SHIPPED RULE	SPECTRA STEP
HGT10134 55.wav	confident real	0.0024 [§]	0.0028 [§]	no
HGT13100 23.wav	confident fake	0.9995 [§]	0.9994 [§]	no
HGT37396 24.wav	Spectra suppressed a likely false alarm	0.0312 [§]	0.0305 [§]	yes
HGT10469 47.wav	deep detectors disagree; drift flagged	0.0027 [§]	0.0348 [§]	no (Spectra never promotes)
HGT32378 68.wav	fake with a stable mains hum	0.9347 [§]	0.8973 [§]	no
HGT17941 58.wav	real with editing seams	0.0049 [§]	0.0097 [§]	no
HGT20801 20.wav	21% [§] voiced, near the gate	0.9640 [§]	0.9384 [§]	no

FILE	SITUATION	P UNDER THE PREVIOUS RULE	P UNDER THE SHIPPED RULE	SPECTRA STEP
HGT23053 93.wav	genuinely uncertain; click flagged	0.4714 [§]	0.4656 [§]	no (Spectra never promotes)

Two rules can change a score, and each is visible in the log when it does: Spectra suppression fired on one of the eight files and on no other; the gate fired on none. Six detectors contributed evidence without touching the score, and the log says so each time. The rejected equal-weight rule would have moved four of the eight files across 0.5[§], all by letting a detector with undisclosed training data vote in full.

Created by Nathan, Hrushi & Tharakesh.

Generated from docs/site/source/pages/how-it-decides.frag f500c6e176a9, README.md 937c075556d3, docs/reports/2026-09-26_worked-examples.md 7ed526e71da6. The markdown is the record. [About this site.](#)

THE METRIC, AND WHAT TO ANSWER WHEN YOU CANNOT TELL

Sixty percent of the grade is one number: the normalized minimum detection cost on the sponsor's test set. Under it, calling a real recording synthetic costs about nine times as much as letting a fake through, and only the order of our scores matters. Those two facts explain most of what this system does, including where it puts the files it cannot judge.

ON THIS PAGE

- 01 The cost function we are judged on
- 02 Try it
- 03 Where the weights come from
- 04 Why only ranking matters
- 05 Why the operating threshold sits far above the midpoint
- 06 Two readings of the same score
- 07 The default answer: the pinned block
- 08 Reading the draft review

THE COST FUNCTION WE ARE JUDGED ON

$$\text{minDCF} = 9.33 \times P_{\text{FA}} + 1 \times P_{\text{miss}}$$

One real file called synthetic costs as much as 9.33 missed fakes. That single ratio shaped every decision in this system.

P_{FA} false-alarm rate: the share of real files scored above the threshold

P_{miss} miss rate: the share of synthetic files scored below it

min the best threshold is chosen by the scorer, so only the ranking of scores matters

Built from the sponsor's weights $C_{\text{FA}} = 4$, $C_{\text{miss}} = 1$ and a prior of 0.3 synthetic (about 0.7 of files are real): $\text{cost} = 4 \cdot P_{\text{FA}} \cdot 0.7 + 1 \cdot P_{\text{miss}} \cdot 0.3 = 2.8 \cdot P_{\text{FA}} + 0.3 \cdot P_{\text{miss}}$, divided by the best constant decision, 0.3, so that 1.0 means "no better than always saying real" and 0.0 is perfect. §

TRY IT

Move the two rates and watch the cost. The arithmetic is exactly the site's formula: $(4 \cdot \text{pfa} \cdot (1 - 0.3) + 1 \cdot \text{pmiss} \cdot 0.3) / 0.3$ with pfa and pmiss as fractions.

False-alarm rate **P_{FA}** 5%



Miss rate **P_{miss}** 20%



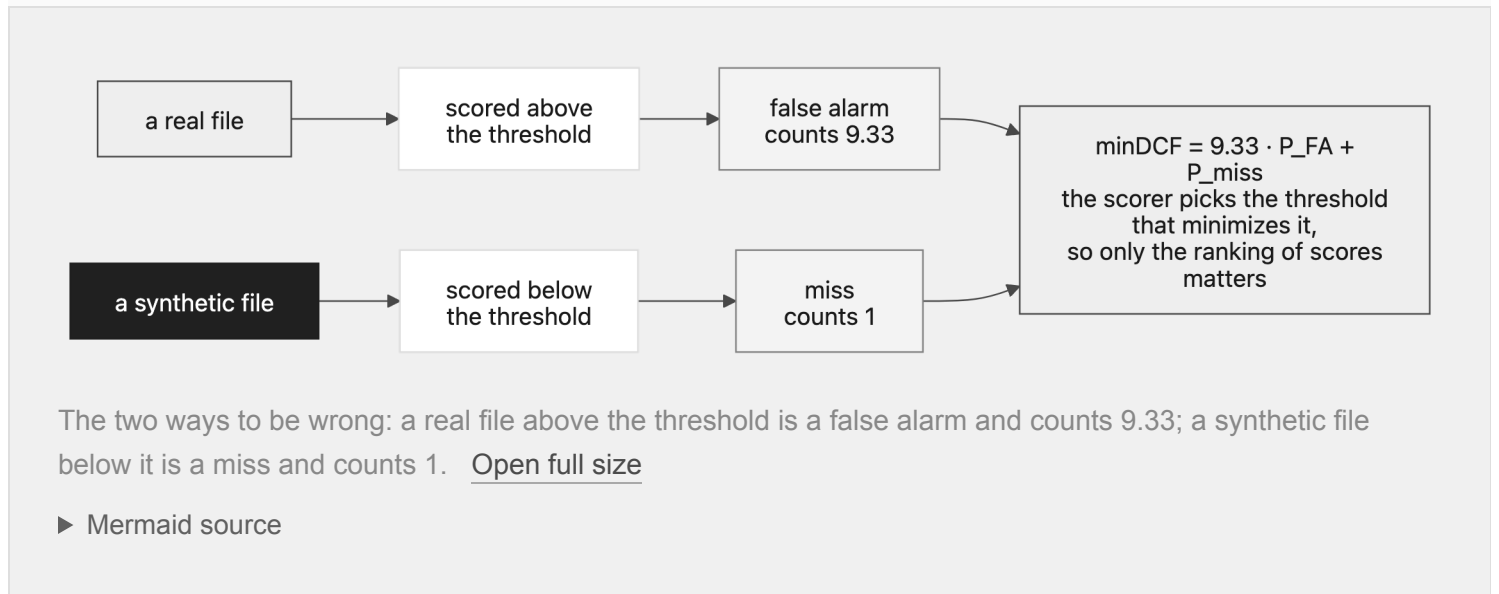
Normalized cost: **0.667**

(1.0 would be the constant "always real" answer)

WHERE THE WEIGHTS COME FROM

The sponsor's instructions set the analyst's scenario: a false alarm (a real recording flagged as synthetic) costs $4 \times$ § a miss, and about 70% § of the test files are real. Our metric module encodes exactly that:

$C_{FA} = 4$, $C_{miss} = 1$, $\pi_{synth} = 0.3$, and the normalized cost is $9.33 \cdot P_{FA} + P_{miss}$. The scorer sweeps every threshold and keeps the smallest cost, so the number is the best the ranking could do with a perfect threshold.



WHY ONLY RANKING MATTERS

Because the scorer chooses the threshold, any monotone rescaling of our scores changes nothing: a [Platt map](#), a z-score, or the cut at $0.5^{\S}$ all leave [minDCF](#) exactly where it was. That is why fusion works in ranks, why the [test share](#) above $0.5^{\S}$ is a smoke alarm and never a selection signal, and why every rule was chosen on minDCF with [EER](#) only reported. [Source: the README's validation section.](#)

WHY THE OPERATING THRESHOLD SITS FAR ABOVE THE MIDPOINT

With a false alarm that expensive, calling a file synthetic only pays when the evidence is strong. On a calibrated probability scale the [Bayes threshold](#) sits near $0.903^{\S}$ under the brief's cost, not at $0.5^{\S}$. That

figure is the cut on a prior-neutral likelihood ratio read as a probability. The probability the TSV carries already includes the prior, and its Bayes cut is 0.8[§]; our metric module's decision logit subtracts the cost term as well, so on it a fixed 50%[§] cut is the Bayes decision. Every intuition anchored on "above 0.5[§] means fake" is off by an order of magnitude in likelihood ratio. This is also why the share of test files above 0.5[§] sitting near the stated prior is reassuring but not a score.

TWO READINGS OF THE SAME SCORE

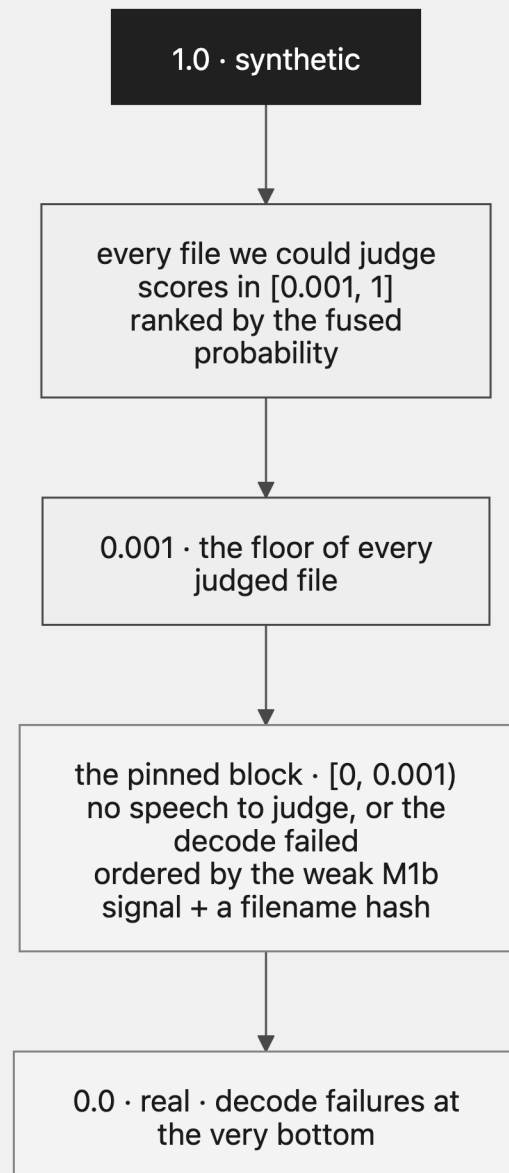
The instructions say 1.0[§] = synthetic. The sponsor's shipped scoring code (ASVspoof5's) treats a higher score as bona fide, the opposite. We follow the instructions and never flip; every model logs its cost under the sponsor's code both ways, and a pre-flipped twin of the draft TSV exists in case NSA's feedback shows they grade inverted. Source: "Score direction" in the README. The second reading also moves the heavy weight: the sponsor code's cost is $P_FA + 4 \cdot P_miss$ at π 0.5, with the 4× on a missed fake. So every readout on this site names its cost: the brief's, or the sponsor code's.

SHIPPED RULE	IN-THE-WILD, BRIEF COST ($9.33 \cdot P_FA + P_MISS$)	IN-THE-WILD, SPONSOR-CODE COST ($P_FA + 4 \cdot P_MISS$)
<u>A3 w 0.2 + E</u>	0.228 [§]	0.239 [§]

THE DEFAULT ANSWER: THE PINNED BLOCK

The brief warns that real clips may be perturbed, that fakes are laundered, and that a naive detector will be fooled. It leaves one question to the team: what do you answer when you cannot tell? Silence, music, static, a file that fails to decode. Our first deep probe scored pure silence at 0.99[§] synthetic, which under this cost would be the most expensive possible mistake if the file were real.

The answer is placement, not a value. Every file we could judge gets a score in $[0.001^{\S}, 1]$. Undecidable files get scores in $[0, 0.001^{\S})$: strictly below all of them, at the real end, ordered inside the block by the weak M1b signal plus a hash of the filename so no two share a value, with decode failures at the very bottom. Source: "The abstention path".



The score line: synthetic at the top, every judged file in $[0.001, 1]$, the pinned block below 0.001 at the real end.

[Open full size](#)

► Mermaid source

Right under both readings of the score

The block is the one hedge that does not depend on which way NSA's code reads our scores. Let q be the fraction of truly synthetic files inside the block. Placing the block at the bottom is optimal under the brief's cost when q is below 80%[§], and under the scoring code's inverted cost when q is above 9.7%[§]. Undecidable files should sit near the 30%[§] base rate, which leaves roughly a 3× margin on each side. Source: consult item 5.

READING OF THE SCORE	WHERE THE EXPENSIVE ERROR SITS	BOTTOM PLACEMENT IS OPTIMAL WHEN	EXPECTED Q FOR UNDECIDABLE FILES
The brief (1.0 [§] = synthetic, false alarm ×4)	a real file ranked high	$q < 80\%^{\S}$	near the 30% [§] prior
The sponsor's code (higher = bona fide, missed fake ×4)	a fake file ranked high	$q > 9.7\%^{\S}$	

No test file is gated today, so the block costs nothing on the current ranking. It only protects against a silent or musical file being scored as synthetic, which our deep models do on their own.

READING THE DRAFT REVIEW

NSA scored one early submission per team. We sent the exact artifact we ship and decided in advance what each returned number would mean. NSA returned minDCF and EER. Source: the README's decoding table and result.

RETURNED MINDCF	WHAT IT MEANS	WHAT WE DO
0.00 [§] –0.20 [§]	our direction; the test set behaves like our holdout	ship unchanged

RETURNED MINDCF	WHAT IT MEANS	WHAT WE DO
0.20 [§] –0.45 [§]	our direction; the test set is more like In-the-Wild	switch to the most false-alarm-robust candidate
0.45 [§] –0.90 [§]	ambiguous	don't flip; fall back to M1b alone
0.95 [§] –1.00 [§]	NSA's code reads our scores inverted	ship the pre-flipped twin

NSA returned minDCF 0.0733[§] and EER 3.534%[§]. The minDCF fell in the pre-declared 0.00–0.20[§] band: our score direction was right, and we kept the submitted rule unchanged.

WHAT WORKED AND WHAT DID NOT

The honest log. Each card names what we did, what it did, and the mechanism behind the number. The negatives are here on purpose: five shortcuts in the data, an equal-weight fusion our holdout loved and we rejected, a fine-tune that failed its gate, four detectors that cannot work on this test set, a codec hunt that came back empty, and an estimate we published and then withdrew.

WHAT WORKED

Band-matching the training audio to the test set

The test set is low-passed at about 7.2 kHz[§] and no training corpus is. A 71-tap Kaiser low-pass at 7.25 kHz[§] on every clip, train and test, took the handcrafted detector from calling 90%[§] of test files synthetic to 42%[§]. Without it, everything above 7 kHz was a train-versus-test fingerprint.

Real-speaker diversity, once band-matched

Adding 40 VCTK speakers (M1b) improved the holdout from 0.159[§] to 0.072[§] and In-the-Wild from 0.380[§] to 0.342[§]. Before band matching, the same addition made In-the-Wild *worse* (0.374[§] → 0.405[§]): the wall was hiding the gain.

Length parity

Gradient-boosted trees over logistic regression

Test clips are about 3.4 s[§] and training clips 5–9 s. Embedding each clip at its own length and training on crops drawn from the test-length distribution removed a repeat-padding seam that only test clips had; it was worth 0.05[§] on its own for the handcrafted detector (0.709[§] → 0.661[§]).

For the handcrafted features: holdout 0.374[§] vs 0.661[§] on the same features. The trees transfer to unseen generators and the linear model does not.

New feature families that see new generators

CQCC frame-to-frame change and LFCC upper-coefficient variability made pro_diff visible (1.00[§] → 0.32[§] out of fold) and cut inner minDCF from 0.614[§] to 0.434[§].

Augmented twins for the handcrafted model

Codec round-trips and random spectral tilt, added as fit-only extra rows, cut holdout-real false alarms from 9.1%[§] to 6.3%[§] and In-the-Wild false alarms from 0.65%[§] to 0.40%[§], and improved the holdout (0.170[§] → 0.137[§]). The price was inner minDCF (0.418[§] → 0.469[§]) and pro_diff (0.30[§] → 0.63[§]). With a false alarm worth 9.33[§] misses, we took the trade. The model is trained with augmented twins and validated on clean rows only.

Treating the container as routing, not evidence

We measured it before using it and found it was the label (below), so the detector never learns from it.

Laundering both classes for compression forensics

Re-encoding a random half of *both* classes through MP3/AAC meant the detector could not learn "MP3 = ElevenLabs/PlayHT". It then detects codec history well (AUC 0.90[§]–0.94[§]).

A pre-declared fusion sweep

Writing the candidates and the selection rule down first stopped us from shipping the equal-weight rule our holdout liked best.

M5 as a minority vote

Alone it lost to the plain probe on every holdout readout. Given 20%[§] of the rank blend beside M1b and the handcrafted model, it cut the fused holdout from 0.014[§] to 0.0065[§] and In-the-Wild from 0.260[§] to 0.228[§]. It errs on different files than M1b, and that is what fusion uses.

Spectra-AASIST as a one-way vote

As a false-alarm suppressor it moved In-the-Wild from 0.322[§] to 0.260[§] at no cost elsewhere. Used that way, if it is wrong about a file the cost is a miss, never a false alarm. The later leakage probes found the step fires only on real clips (4–5%[§] of them) and never on a fake, under every perturbation we tried.

The phase cues as evidence

Peak phase-versus-magnitude coherence separates grad_tts from real speech (AUC 0.80[§]), is corpus-neutral, and names a vocoder property a person can check.

WHAT DID NOT WORK

Five shortcuts in the data, all found and closed

The codec hunt came back empty

Level: test clips peak near full scale and training real speech near 0.5[§]; peak alone predicted the label (AUC 0.66[§]). Closed by normalizing each input. Leading silence: LibriSpeech starts with about 0.37 s[§] of silence, test clips 0.06 s[§]. Closed by trimming both. Tiling: repeat-padding short test clips put a seam only in test data. Closed by length parity. Container: every test file is 16 kHz PCM WAV with the same FFmpeg tag. Closed by one decode path. The 7.2 kHz[§] wall: closed by band matching.	We checked whether the wall is a codec, to reproduce it more faithfully in training. It isn't: across 34 MP3 and AAC settings none matched it better than our own Kaiser low-pass (the closest was 3%[§] nearer on distance and worse on spectral holes). It is a smooth low-pass, like a resampler's anti-alias filter, so the planned symmetric codec refit never ran.
The container is the label LibriSpeech is FLAC, LJ is 22 kHz WAV, and DiffSSD's ElevenLabs and PlayHT are MP3. PlayHT's MP3s carry the exact encoder tag that all 1,671[§] test files carry. A learned metadata model would score perfectly on our holdout for the wrong reason, then call the whole test set PlayHT. The container detector returns a constant 0.5[§] on this test set by design.	Compression forensics sees the pipeline, not the class With codec history equalized, class holdout minDCF is 0.90[§], and it calls 81%[§] of the test set synthetic, because the test set reads as laundered throughout (mean P(laundered) 0.62[§]). It stays as evidence.
ENF cannot be validated here Stable mains hum is in 36%[§] of LJ Speech clips (a home studio) and 21%[§] of LibriSpeech, and in almost no generated audio, so a learned version would learn "hum = LJ = real". Only 24 test files carry hum. Rule-based and mild on purpose; evidence only.	Splice finds vocoder artifacts, not edits On training data the seams are single-sample clicks in 27%[§] of WaveGrad2 and DC jumps in 9–13%[§] of several generators <i>and</i> 9%[§] of LibriSpeech. The sponsor said no test clip is partially synthetic. It flags 58 test files; evidence only.

Speaker drift points the wrong way

Fakes are the *most* self-consistent voices: mean window-to-window similarity separates them toward fake at AUC 0.73[§]. A cloned voice doesn't vary from second to second the way a person does. Drift flags 16%[§] of test files, mostly the hard real recordings. Fused, it would push those toward synthetic, so it is evidence only.

Fine-tuning XLS-R lost to a frozen probe

Fine-tuning all 12 kept layers fit the seen generators and transferred worse to the unseen commercial one: 0.38–0.41[§] on the ElevenLabs fold vs 0.31[§] with the backbone frozen. The frozen-backbone head then lost to the plain probe on the holdout (0.363[§] vs 0.159[§]), on WaveGrad2, on LibriSpeech real speech and on clips under 4 s (0.60[§]). It won only on In-the-Wild false alarms (0.45%[§] vs 0.7%[§]). Its one use is as a 20%[§] voice in fusion. Twelve A100 instances, \$5.42[§] in total.

Pruning corpus-cue columns hurt every time

83 handcrafted columns separate LJ from LibriSpeech more than they separate real from fake. Dropping them made every readout worse, including LibriSpeech false alarms (0.54[§] → 0.59–0.61[§]); stripping v3's own cue columns took it from 0.614[§] to 0.848[§]. The check told us which per-source numbers to read, not which columns to delete.

The handcrafted detector's test-domain offset

Even band-matched, the test recordings have a darker, codec-like spectral envelope. The 234[§] columns tell train from test at AUC 0.99[§], and 84%[§] of the shifted columns move toward "fake". Pruning the shifted columns made it worse (the rest still separate at 0.98[§]); augmentation removed only part of it. This is part of why it gets 20%[§] of the vote, not half.

grad_tts is invisible to the handcrafted model under our

Prosody beyond pitch

validation Its phase cue is shown by no other generator, so when grad_tts is the held-out one, nothing teaches the cue: 1.00[§] in every variant. That is an honest limit of holding out by generator.	Pause fraction and count, voiced fraction, loudness dynamics and zero-crossing rate all sat within 0.05[§] of chance.
Equal-weight fusion: rejected Our first fusion averaged the two detectors equally. On the holdout it looked like the best thing we had: 0.018[§] and 0.015[§], against 0.072[§] for M1b alone. On In-the-Wild it doubled the misses, from 28%[§] for M1b alone to 58–60%[§], and pushed the share of test files called synthetic up to 32–40%[§]. The holdout gain was below the holdout's resolution floor; the In-the-Wild loss was large and consistent. The handcrafted column is excellent on generators it has seen and blind to web-sourced fakes (97%[§] missed on its own), so giving it half the vote traded real-world detection for a holdout number.	Spectra kept out of the vote, though a full vote scores better With M5 in the blend, fusing everything equally beats the shipped rule on every labeled readout (holdout 0.0045[§], In-the-Wild 0.214[§] and 0.196[§]). It does so by giving Spectra-AASIST a full vote. We ruled that out before seeing these numbers, because Spectra's training data is undisclosed and our validation rows may be in its training set. The choice rests on that argument, not on these readouts.
Our holdout overstates real-world skill Every XLS-R probe we trained sits at 0.34–0.41[§] on In-the-Wild against 0.07–0.16[§] on the holdout, and the handcrafted model gets 1.00[§] there. Channel robustness, not more read speech or more fake data, is the gap we did not close. A classifier separating our holdout's real speech from In-the-Wild's, run over the test	A pretrained model we can't audit, then probed Spectra-AASIST separates our inner rows perfectly (AUC 1.000[§]). Its model card names no training set, so its inner, holdout and In-the-Wild numbers may all be in-sample. We probed for the symptoms of memorisation instead: on 500[§] holdout clips under four label-preserving changes it

files, puts the wild share at $\hat{\lambda} = 0.51$ [§] (95% [§] CI 0.12–0.64 [§]). The test set is partly wild, so neither validation set represents it alone; the shipped rule beats the previous one on both, so the choice does not depend on λ.	was the most stable detector (mean ΔAUC 0.0008 [§], against 0.0041 [§] for M1b); on 572 [§] MLAAD clips from 143 [§] generators it had the lowest miss rate (14.9% [§], against 22.7% [§] for M1b, 28.7% [§] for M5, 56% [§] for the handcrafted model); its suppression step never raised the fused holdout cost. Verdict: keep it as a suppressor only. The one number we still can't trust is its In-the-Wild score, because its authors evaluated on that set.
Peak-normalizing Spectra's input A fix for its within-fake level correlation cost 0.033 [§] on real speech and did not remove the correlation, so the level was a proxy, not the cue. Rejected.	Additive noise is the shipped rule's weak spot At 20 dB SNR white noise on 500 [§] holdout clips the handcrafted model loses almost all discrimination (AUC 0.998 [§] $\rightarrow$ 0.64 [§], minDCF 0.048 [§] $\rightarrow$ 0.996 [§]); M1b rises from 0.056 [§] to 0.271 [§], and the shipped rule from 0.012 [§] to 0.269 [§]. Spectra holds at 0.116 [§], but noise pushes it out of its "strongly real" region, so its suppression step never fires. Found after the rule was frozen and not fixed; the pointer is noise augmentation of the handcrafted training, applied to both classes.
A wild real-speech corpus, planned and not attempted It was the fusion consult's suggested fix for the In-the-Wild gap. We dropped it at 12:16 on Saturday because a download, extraction and	Our first estimate of how wild the test set is was wrong

refit would not fit the time box, and the time went to the Spectra probes and the λ estimate. The two candidates looked unusable anyway: VoxCeleb1's test set needs a request form, and SpoofCeleb's real speech has been speech-enhanced.	It said $\hat{\lambda} = 0.947$ with a classifier AUC of 0.99, and read "wild" as "codec-processed". One feature did the work: a whole-spectrogram noise-floor percentile that landed in the 7–8 kHz stopband. It was measuring the test files' double low-pass, not the recording channel: one MP3 round-trip moved 91% of our clean clips to "wild". Dropping that feature, under the rule we had already declared for columns above 7 kHz, gave 0.51 (MP3 control 91% $\rightarrow$ 9%). The same kind of shortcut as the 7.2 kHz wall, read through a feature that didn't look like a band feature.
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Created by Nathan, Hrushi & Tharakesh.

Generated from docs/site/source/pages/what-worked.frag ec275a34bd4f, README.md 937c075556d3. The markdown is the record. [About this site.](#)

THE TEN DETECTORS

Ten detectors run on every file, all behind one contract: a score in [0, 1] that rises with synthetic likelihood, one plain-English sentence of evidence, named features, and a status. Three vote, one may only lower the score, one gates, five are evidence and routing. Each card names the sponsor's rubric technique it implements, what it measures, its role, its numbers and one real sentence it wrote about a test file.

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- 01 The rubric, mapped
- 02 The three that vote
- 03 The one that may only lower a score
- 04 Evidence and routing, never scored
- 05 The gate
- 06 All ten in one table

THE RUBRIC, MAPPED

The sponsor awards diversity points for each distinct technique a system actually uses, plus a bonus for choosing analyses per file. Source: the README's technique table.

RUBRIC TECHNIQUE	DETECTOR	HOW IT IS USED
Container, digital file forensics and metadata	<u>container</u>	routing
Spectral / frequency-domain analysis	<u>handcrafted</u> , <u>Spectra-AASIST</u>	fused (20% [§]); suppressor
Prosody and phonetics	<u>handcrafted</u>	fused
Acoustic-environment consistency (ENF)	<u>enf</u>	evidence
Compression forensics	<u>compression</u>	evidence
Speaker-embedding consistency	<u>speaker_drift</u>	evidence
Deep-learning anti-spoofing	<u>M1b</u> , <u>M5</u> , <u>Spectra-AASIST</u>	fused (Spectra as suppressor only)
Splice / discontinuity detection	<u>splice</u>	evidence
Agentic orchestration bonus	the <u>router</u> and the <u>speech gate</u>	rules over measured facts, a routing log per file, and a measured router-on-versus-off ablation

THE THREE THAT VOTE

FUSED · WEIGHT 0.6 §

M1b: frozen XLS-R with a logistic probe

Technique: deep-learning anti-spoofing. **What it measures:** the hidden state of layer 7 of XLS-R (frozen), averaged over time, then a class-balanced logistic regression and a Platt map. M1b adds 40 VCTK real speakers from ASVspoof 2019 and a spoof anchor to the training folds only. Layer 7 won inner cross-validation; mid-depth layers carry the acoustic detail that separates vocoders, the top layers are tuned for phonetic content.

Numbers: inner 0.301 §, holdout 0.072 § (EER 1.4% §), In-the-Wild 0.342 § (the results table rounds it 0.343 §), In-the-Wild false alarms 0.8% §, test share above 0.5 § 26.8% §.

XLS-R layer-7 probe: calibrated log-likelihood ratio -5.78 (real-like, P=0.00)

Sentence from the README's trace of HGT1046947.wav. Spec: the XLS-R probe.

FUSED · WEIGHT 0.2 §

Handcrafted spectral and prosody features

Techniques: spectral analysis and prosody. **What it measures:** 234 § features: spectral contrast, flatness, flux, roll-off, MFCC, LFCC and CQCC statistics, group-delay and phase-coherence cues; YIN pitch spread and slope, jitter and shimmer, loudness modulation, pauses, breath frames; gradient-boosted trees evaluated in numpy. Pitch variability is the one prosody cue that carried signal. The shipped version (v5b) is trained with augmented twins and validated on clean rows only.

Numbers: inner 0.469 §, holdout 0.137 § (EER 3.1% §), In-the-Wild false alarms 0.40% §, test share above 0.5 § 45.1% §. On its own it is blind to web-sourced fakes; that is why it gets a fifth of the vote, not half.

spectral/prosody features real-like (P=0.16): LFCC 16 frame-to-frame change 3.0 SD below real speech (toward synthetic); spectral contrast, band 0 0.8 SD above real speech (toward real); LFCC 1 variability 1.1 SD below real speech (toward real)

Sentence from worked example 4. Spec: handcrafted features.

FUSED · WEIGHT 0.2 §

M5: a trained head on XLS-R

Technique: deep-learning anti-spoofing. **What it measures:** the first 12 layers of XLS-R with learned layer weights, attentive statistics pooling and a linear head, trained on rented A100s with class-blind augmentation (noise, telephony band-limits, reverb, codecs, RawBoost). It did not clear its gate as a replacement for the probe, but it is wrong on different files, and as a third rank input it passed the second pre-declared sweep.

Numbers: inner 0.302 §, holdout 0.363 § (EER 6.0% §), In-the-Wild false alarms 0.45% §, test share above 0.5 § 25.8% §.

XLS-R fine-tuned head (M5, 12 layers, attentive pooling): logit +5.64 (synthetic-like, P=1.00)

Sentence from the README's trace. Spec: the M5 head.

THE ONE THAT MAY ONLY LOWER A SCORE

SUPPRESSOR · WEIGHT 0

Spectra-AASIST (M3)

Techniques: deep-learning anti-spoofing, spectral. **What it measures:** an off-the-shelf anti-spoofing model (its own XLS-R plus a graph-attention back end), run with no training through the same band-matched input path and crops; short clips are zero-padded to one window. Output: a spoof-minus-bonafide margin. **Role:** if the margin is below -3 and the fused rank is above 0.5 §, the rank is halved. It can never raise a score.

Numbers, all possibly in-sample because its model card names no training set: inner 0.0045[§], holdout 0.012[§] (EER 0.16%[§]), In-the-Wild 0.065[§] (its authors evaluated on that set), In-the-Wild false alarms 0.0%[§], test share above 0.5[§] 29.3%[§]. The leakage probes support the suppressor role: the most perturbation-stable detector, the lowest miss rate on unseen generators, and a step that never raised the fused cost.

Spectra-AASIST: spoof-minus-bonafide margin +9.50 (synthetic-like, P=1.00)

Sentence from the README's trace. Spec: Spectra-AASIST.

EVIDENCE AND ROUTING, NEVER SCORED

EVIDENCE

Compression forensics

Technique: compression forensics. **What it measures:** 19[§] codec-trace features in 3–7 kHz (spectral holes, floor depth, effective bandwidth), trained with both classes laundered through MP3/AAC so it cannot learn "MP3 = fake". It detects codec history well and the class barely, because the test set reads as laundered throughout. Its effective-bandwidth reading is a routing fact.

Numbers: inner 0.931[§], holdout 0.900[§] (EER 20.1%[§]), test share above 0.5[§] 80.7%[§].

compression-trace features synthetic-like (P=0.97): whole-clip floor depth 1.7 SD below real speech (toward synthetic); 3–7 kHz valley depth 0.7 SD below real speech (toward synthetic); deep-hole flicker frame to frame 1.7 SD above real speech (toward synthetic)

Sentence from worked example 1, a confident real file. Spec: compression forensics.

Container and metadata

Technique: container, file forensics and metadata. **What it measures:** header facts (codec, rate, bit depth, encoder tag). Rule-based on purpose: on our training data the container *is* the label, and every test file carries the same FFmpeg tag, so it returns a constant score and routes compression forensics. It never reads filenames or filesystem timestamps, because git, zip and Docker rewrite them.

Numbers: none by design (a constant 0.5[§] on this test set).

```
wav/pcm_s16le 16000 Hz mono 16-bit, 1 tag(s) [encoder=Lavf58.29.100]: no class
evidence in the container; written by FFmpeg (libavformat), so the file was re-muxed
or transcoded at least once
```

The sentence every one of the eight files received. Spec: container.

ENF: mains hum

Technique: acoustic-environment consistency. **What it measures:** a 50 or 60 Hz hum and its stability on short windows. Rule-based and mild on purpose: stable hum is in a third of LJ Speech clips and almost no generated audio, so a learned version would learn "hum = real".

Numbers: flags 24[§] test files; never moves a score.

```
no mains hum at 50 or 60 Hz (best 60 Hz candidate: median SNR 2.8 dB, above 12 dB in
0% of frames): no environment evidence either way
```

```
enf: mains hum present, stable (recording-environment evidence)
```

First sentence from worked example 1; the routing-log line from example 5, a fake with a stable hum. Spec: ENF.

EVIDENCE

Splice and discontinuities

Technique: splice / discontinuity detection. **What it measures:** sample-level clicks and DC-offset jumps between 100 ms[§] windows. On this data it finds vocoder artifacts more than edits, and the sponsor said no test clip is partially synthetic, so it is evidence.

Numbers: flags 58[§] test files; never moves a score.

editing seam at 1.59 s: 1 click(s) (largest step 17x the surrounding waveform); consistent with a cut or inserted segment (background-floor range 26 dB, informational)

Sentence from worked example 8. Spec: splice.

EVIDENCE

Speaker drift

Technique: speaker-embedding consistency. **What it measures:** ECAPA embeddings on one-second windows; flags a voice that changes within the clip. The finding that keeps it out of the score: fakes are the *most* self-consistent voices, so fused it would push hard real recordings toward synthetic.

Numbers: the mean window-to-window similarity separates classes toward fake at AUC 0.73[§]; drift flags 16%[§] of test files.

voice drifts across the clip: minimum window-to-window speaker similarity -0.10 over 14 windows

Sentence from the README's trace. Spec: speaker drift.

THE GATE

GATE

Speech gate

Part of the orchestration. What it measures: whether a file contains speech at all, from five cues: not silent, at least 5%[§] voiced, pitch moving, loudness moving, not spectrally flat. Each threshold was set beyond the extreme value in the full test set, so it only catches silence, tones, static and noise; a file that fails skips fusion and goes to the pinned block.

Numbers: gates 0 of 1,671[§] test files.

speech present: voiced 28% of frames, pitch spread 0.34, loudness std 18.5 dB

Sentence from the README's trace. Spec: the speech gate and the default-answer policy.

ALL TEN IN ONE TABLE

Normalized minDCF unless marked; "inner" is pooled out-of-fold over the inner rows; "ITW FA" is the In-the-Wild real-speech false-alarm rate at the model's own inner-fold threshold; "test > 0.5[§]" is a sanity check, not a score. Source: the README's Numbers section.

DETECTOR	ROLE	INNER	HOLD OUT	HOLDOUT EER	ITW	ITW FA	TEST > 0.5 [§]
M1b v3	fused 0.6 [§]	0.30 1 [§]	0.07 2 [§]	1.4% [§]	0.34 2 [§]	0.8% §	26.8% §

DETECTOR	ROLE	INNER	HOLD OUT	HOLDOUT EER	ITW	ITW FA	TEST > 0.5 [§]
Handcrafted v5b	fused 0.2 [§]	0.46 9 [§]	0.13 7 [§]	3.1% [§]	—	0.40 % [§]	45.1% §
M5 head	fused 0.2 [§]	0.30 2 [§]	0.36 3 [§]	6.0% [§]	—	0.45 % [§]	25.8% §
Spectra-AASIST (possibly in-sample)	suppress or	0.00 45 [§]	0.01 2 [§]	0.16% §	0.06 5 [§]	0.0% §	29.3% §
Compression	evidence	0.93 1 [§]	0.90 0 [§]	20.1% §	—	—	80.7% §
Container, ENF, splice (rule-based)	routing, evidence	1.0 [§]	1.0 [§]	—	—	—	—
Speaker drift	evidence	not scored as a class detector; see the card above					
Speech gate	gate	no score; gates 0 of 1,671 [§] test files					

Created by Nathan, Hrushi & Tharakesh.

Generated from docs/site/source/pages/detectors.frag 66a59ef5d34c, README.md 937c075556d3, docs/reports/2026-09-26_worked-examples.md 7ed526e71da6. The markdown is the record. [About this site](#).

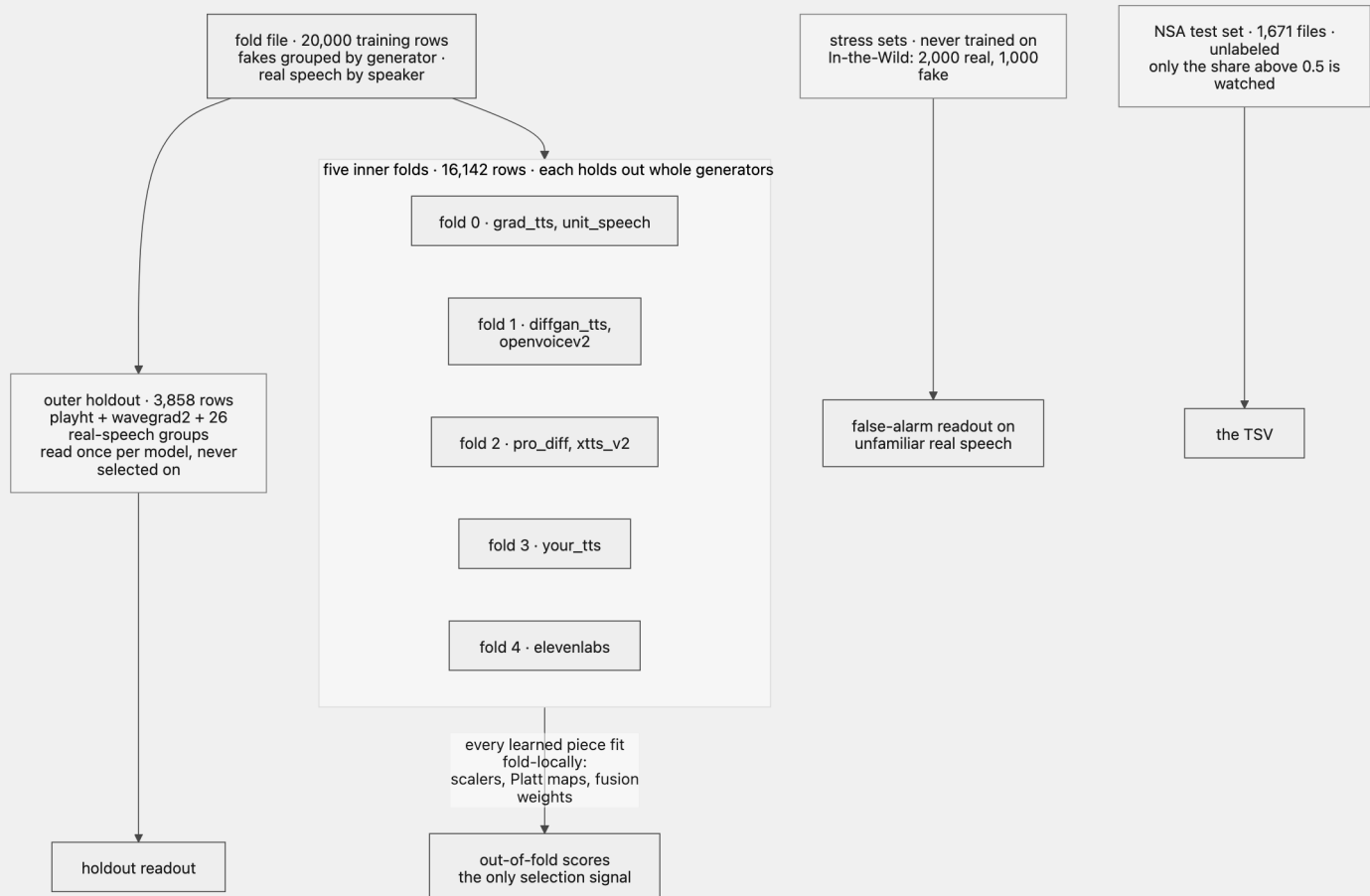
VALIDATION: HOW WE KNOW WHAT WE KNOW

Every number on this site comes from a split that holds out whole generators and whole speakers, never single clips. Selection uses only out-of-fold scores; the holdout is read once per model; In-the-Wild is never trained on; the test set is unlabeled and only its share above 0.5^{\$} is watched. The fusion rule was chosen by rules written down before the results existed.

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- 01 Folds by generator and speaker, never by clip
- 02 The pre-declared sweeps
- 03 The day, in order
- 04 The draft review

FOLDS BY GENERATOR AND SPEAKER, NEVER BY CLIP



The fold scheme: 20,000 training rows split into five inner folds that each hold out whole generators, one outer holdout read once, stress sets never trained on, and the unlabeled test set. [Open full size](#)

► Mermaid source

- **The fold file** has 20,000[§] rows: 10 DiffSSD generators, LJ Speech and LibriSpeech. Fakes are grouped by generator; real speech by speaker (LJ, a single speaker, by chapter). Five inner folds each hold out whole generators: grad_tts + unit_speech, diffgan_tts + openvoicev2, pro_diff + xtts_v2, your_tts, ElevenLabs. Every learned piece of a detector, scalers and Platt maps included, is fit fold-locally, and selection uses only inner out-of-fold scores.
- **One outer holdout, read once per model:** PlayHT and WaveGrad2 plus 26 real-speech groups (3,858[§] rows). Nothing was ever trained, tuned or selected on it. Extra training data (ASVspoof 2019 real speakers, MLAAD, the rest of DiffSSD) enters inner folds only, so holdout numbers stay comparable across every model.
- **The holdout has a resolution floor.** With two held-out generators its noise is about ± 0.07 – 0.10 [§], so we treat any gap under 0.15 [§] as noise. It can tell a good detector from a bad one, not a good fusion rule from a slightly better one.

- **In-the-Wild is the stress test:** 3,000[§] clips (2,000[§] real from 54 speakers, 1,000[§] fake), evaluation only, never trained on. For each model we report minDCF and the false-alarm rate at the threshold chosen on inner folds, under both cost weightings.
- **The test set as a smoke alarm.** We track the share of test files scored above 0.5[§] against the ~30%[§] the brief states. It flagged the band mismatch. It is never a selection signal: any monotone rescaling changes it without changing minDCF.

Source: the README's validation section.

THE PRE-DECLARED SWEEPS

The fusion rule got where it is in two steps, each written down before any result was run. Source: "The fusion rule" in the README and the sweep report.

Step 1, frozen at 08:13

- **Candidates:** a rank blend $(1-\alpha) \cdot \text{rank}(\text{M1b}) + \alpha \cdot \text{rank}(\text{handcrafted})$ for α from 0 to 0.5[§]; min and max rules; a cascade; a non-negative stacker shrunk toward equal weights. Then, on top of the winner, Spectra as a false-alarm suppressor.
- **Rule:** keep candidates within 0.03[§] of the best inner out-of-fold score whose In-the-Wild minDCF under the sponsor code's cost is ≤ 0.45 [§]. Among those, take the best In-the-Wild minDCF under the brief's cost, with ties going to more M1b. Add Spectra suppression only if it helps In-the-Wild by ≥ 0.01 [§] without hurting inner or holdout by more than 0.01[§]. No iteration afterward.
- **Result:** $\alpha = 0.2$ [§] (tied with 0.3[§]; the tie went to more M1b). Adding Spectra suppression moved In-the-Wild from 0.322[§] to 0.260[§] and hurt nothing, so it was applied. Final: inner 0.140[§], holdout 0.014[§], In-the-Wild 0.260[§] (0.267[§] under the sponsor code's cost).

Step 2, the M5 addendum, ratified at noon

- **Written at 08:35, before any run:** two M5 candidates and a stricter replacement rule. Replace the frozen rule only if In-the-Wild improves by ≥ 0.01 [§] under the brief's cost, gets no more than 0.01[§] worse under the sponsor code's, the holdout gets no more than 0.05[§] worse, and inner no more than

0.03[§] worse. The candidates: M5 as a second false-alarm suppressor (F) and M5 as a third rank input (A3).

- F missed the bar: its In-the-Wild gain was 0.003[§].
- A3 at weight 0.2[§], plus the same Spectra step, qualified: inner 0.135[§] vs 0.140[§], holdout 0.0065[§] vs 0.014[§], In-the-Wild 0.228[§] / 0.239[§] (brief / sponsor-code cost) vs 0.260[§] / 0.267[§].
- We had predicted A3 would lose on short clips and LibriSpeech real speech. With the Spectra step it improved both.
- On the test set the switch changes little: Spearman 0.974[§] against the previous rule's scores, and 3 of 1,671[§] files cross 0.5[§].

THE DAY, IN ORDER

Times are Saturday, September 26, local to the event. Each entry links to the document that records it; where documents disagree by a few minutes the README's time is shown.

● 05:32

First fusion: the two detectors averaged equally. Best holdout we had; doubled the In-the-Wild misses.

[README](#)

● 08:13

The rule frozen by the pre-declared sweep: $\alpha = 0.2$ [§] with Spectra suppression. [README](#)

● 08:35

Two M5 candidates and a stricter replacement rule written down, before any run. [README](#)

● ~09:15

Addendum results appended; [A3 w 0.2 + E](#) qualifies; F does not. [sweep report](#)

● 09:25

Held: the previous rule stays the draft-review payload; the runner side is built behind a flag. [README](#)

● 12:03

Nathan ratifies the switch to [A3 w 0.2 + E](#). [README](#)

● 12:30

The draft sent to NSA by Discord DM as CrossExam_predictions.tsv. [README](#)

12:35

The runner's default switched to the shipped constants file. [STATUS](#)

13:20

The channel-robustness lane done: λ corrected, the codec hunt negative, Spectra's suppression kept, the noise weakness found. [STATUS](#)

THE DRAFT REVIEW

For the [draft review](#), we sent the exact TSV we ship on Saturday at 12:30 and decided in advance what each returned number would mean. NSA returned minDCF 0.0733[§] and EER 3.534%[§]. The minDCF fell in our pre-declared 0.00–0.20[§] band, so the rule stayed unchanged; the decoding table is on [the metric page](#). [Source: the README](#).

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REPRODUCE IT

The commands below are the README's, verbatim. Data, weights, model directories and submission TSVs are not in the repository; the layout they are expected in is stated, and the parity checks say how closely a live run reproduces the submitted file.

ON THIS PAGE

- 01 Setup
- 02 Score a directory
- 03 Reproduce in Docker (offline, CPU; not a graded deliverable)
- 04 Where each artifact lives
- 05 Repository layout

SETUP

Requires uv and FFmpeg. [Source: the README's setup section.](#)

```
uv sync
cp .env.example .env
uv run pytest -q
uv run ruff check .
```

Data, weights, model directories and submission TSVs are gitignored; only the experiment log is tracked. Expected layout: the NSA test set at `data/nsa/HackGTHearsayTesting/`, the template at

data/nsa/HearsayScoreKey4TeamX.tsv , pretrained weights under weights/ , trained bundles under models/ .

SCORE A DIRECTORY

```
uv run python scripts/run_pipeline.py --in data/nsa/HackGTHearsayTesting --out outputs/run
```

--fusion models/fusion_v2/constants.json selects the shipped rule. It writes CrossExam_predictions.tsv , one explanation JSON per file under results/ , a resumable results.jsonl , timings.csv and run_meta.json . Rerun the same command to resume after a crash. --flip also writes the pre-flipped twin; --rule , --policy and --flip re-fuse the cached logits without reloading a model. Speed with M5 loaded: 1.34 s per file on the full test set (38 minutes wall on a Mac shared with other jobs); peak memory 4.0 GB . [Source: the README.](#)

Parity

CHECK	RESULT
Exported logits → shipped constants → policy, vs the submitted TSV, all 1,671 rows	max abs diff 6.7e-16
Same, --flip , vs the flipped twin	max abs diff 7.2e-16
Live from audio, all files, vs the submitted TSV	Spearman 1.000000 , max 9.3e-4 , mean 1.8e-5 ; 0 verdict flips, 0 scorer errors
M5 logit, live CPU on WAV vs the A100 export on FLAC	max 0.035 , median 7.0e-4

CHECK	RESULT
Handcrafted logit, live vs export	max 4.4e-16 [§] (numpy trees, identical)
M1b logit, live CPU vs the MPS-extracted export	max 5.5e-4 [§]
Spectra-AASIST logit, live CPU vs MPS export	max 1.1e-4 [§]
Previous rule: exported logits → previous constants vs the 08:13 TSV, all rows	max abs diff 1.1e-16 [§]

REPRODUCE IN DOCKER (OFFLINE, CPU; NOT A GRADED DELIVERABLE)

NSA said on Saturday that the image is not required; the deliverable is the TSV and the README. The image is kept as evidence that the pipeline runs offline from a clean build. **The recorded image runs the previous rule, not the shipped one:** it does not contain the M5 checkpoint. [Source: the README.](#)

```
bash docker/build.sh
docker run --network none -v <test_dir>:/data:ro -v <out_dir>:/out -v <path>/HearsayScore
```

The image contains every engineered detector, the M1b probe, Spectra-AASIST, the ECAPA speaker model, the handcrafted bundle and the fusion constants, with a sha manifest of the shipped files checked at start. It refuses to start unless it is offline, and nothing is downloaded at run time. Always pass `HEARSAY_TEAM` : the runner's default team name is not ours. Recorded image `hearsay:20260926-0916` : 50[§] test files scored inside the image against the previous rule's TSV, Spearman 1.0[§], max abs diff 2.02e-4[§]; decoded audio identical to the Mac on 50 of 50 files; the smoke test passed, including the refusal to run online.

WHERE EACH ARTIFACT LIVES

ARTIFACT	PATH
Experiment log (every TSV with its holdout score)	<code>submissions/log.csv</code>
Shipped fusion constants	<code>models/fusion_v2/constants.json</code> ; previous rule <code>models/fusion_v1/constants.json</code>
Per-detector score exports	<code>outputs/detector_scores/<name>.csv</code> (regenerable; commands in each report)
Model bundles and their readouts	<code>models/<run>_<stamp>/meta.json</code>
Submitted TSV	<code>submissions/20260926-0914_M4_sweep_A3_w0.2_E_CANDIDATE_our_direction.tsv</code> (sent as <code>CrossExam_predictions.tsv</code>)

ARTIFACT	PATH
Fold file	<code>splits/nsa_folds.csv</code> (+ <code>nsa_folds_plus_asv19.csv</code> for M1b)
Experiment write-ups	<u><code>docs/reports/</code></u>
Outside review sessions we asked for, and their answers	<u><code>docs/consults/</code></u>
Run specs and audits	<u><code>docs/specs/</code></u>

REPOSITORY LAYOUT

src/hearsay/ tests/ scripts/ docker/ docs/	application code; detectors/ holds the contract and every detector's pytest suite (hermetic; data- and weight-dependent tests are marked slow); data/ holds the data; detectors/ holds the contract and every detector's extraction, training, fusion, runner, cloud and ops scripts build, entrypoint, smoke, parity and asset-manifest scripts plan, architecture, code map, reports/, specs/, consults/, handoff
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submissions/	log.csv (tracked) and submission TSVs (ignored)
data/ weights/ models/	local only
.claude/	Claude Code skills and review scripts shared by the team

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CREDITS AND AI USE

We used AI coding assistants heavily, and we say so plainly. The team decided what to build and why; Claude Code built it under our direction; Codex reviewed plans and audited finished rungs; two strategy questions went to an outside multi-model review. Nothing on the scoring path calls a hosted model.

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- 01 AI use, in short
 - 02 Pretrained models
 - 03 Datasets
 - 04 Frameworks and tools
 - 05 The team
-

AI USE, IN SHORT

- **Claude Code (Anthropic)** wrote most of the code and documents under our direction, in parallel sessions per rung: loader, metrics, fold file, the M1 probe, the engineered detectors, M3 scoring, the M5 cloud pipeline, the fusion sweep, the runner, the API, the Docker image, the documentation, and this site and its generator. We chose the approach, set the gates and selection rules, reviewed results and made every ship decision.
- **OpenAI Codex (Codex CLI)** reviewed plans and audited finished rungs through our plan-review workflow.

- **Outside consults.** Two strategy questions, one on M5's extra data and one on fusion, went to an outside multi-model review (VeriLM; the fusion memo was answered by Claude and Gemini with a synthesis). The prompts and answers are saved in the repository, and we record which advice we adopted and which we ignored: see the [consult records](#).
- **No AI on the scoring path.** No LLM or hosted API runs at inference; the runner loads every model from local files, and the Docker image runs with networking off.

The full disclosure, including which components each tool touched and what we built versus what the AI wrote, component by component, is kept current in [CLAUDE.md](#), "AI use disclosure"; the README's summary is at ["AI use and credits"](#).

PRETRAINED MODELS

MODEL	SOURCE	LICENCE	HOW WE USED IT
XLS-R 300M, facebook/wav2vec2-xls-r-300m	Hugging Face (Meta)	Apache-2.0	Frozen backbone for M1/M1b (layer 7) and M5 (first 12 layers); both are fused
Spectra-AASIST, lab260/Spectra-AASIST	Hugging Face	unclear: repo header says Apache-2.0, model card says MIT	M3, scored off the shelf, false-alarm suppression only
ECAPA-TDNN, speechbrain/spkrec-ecapa-voxceleb	Hugging Face (SpeechBrain)	Apache-2.0	Speaker-drift evidence

MODEL	SOURCE	LICENCE	HOW WE USED IT
WavLM Large / Base, microsoft/wavlm-large , microsoft/wavlm-base	Hugging Face (Microsoft)	no licence on the card; released through Microsoft's unilm repo (MIT)	Downloaded as bake-off challengers; never run

DATASETS

DATASET	SOURCE	LICENCE	HOW WE USED IT
NSA HEARSAY data: test set, resampled LJ subset, DiffSSD, ASVspoof5 scoring code	NSA, provided during the event	event terms	Test set; training sample and fold file; metric reference
LJ Speech 1.1 [§]	keithito.com	public domain	Real speech (one speaker); training
LibriSpeech dev-clean / test-clean	openslr.org/12	CC BY 4.0 [§]	Real speech (multi-speaker); training
ASVspoof 2019 LA	University of Edinburgh	ODC Attribution	Public shakedown of M1; 40 VCTK real speakers and an A01–A06 anchor as training-

DATASET	SOURCE	LICENCE	HOW WE USED IT
	DataShare		only rows for M1b and M5
In-the-Wild (Müller et al. 2022), mueller91/In-The-Wild	Hugging Face	CC-BY-SA-4.0 on Hugging Face; Apache-2.0 on deepfake-total.com	Stress test only; never trained on
MLAAD, mueller91/MLAAD	Hugging Face (gated)	non-commercial notice	6,180 [§] fake clips from 143 [§] TTS models, training-only for M5

Source: the README's [models table](#) and [datasets table](#).

FRAMEWORKS AND TOOLS

PyTorch and torchaudio (BSD), Hugging Face transformers and huggingface_hub (Apache-2.0), SpeechBrain (Apache-2.0), librosa (ISC), soundfile (BSD-3-Clause), NumPy, SciPy, scikit-learn and pandas (BSD-3-Clause), LightGBM (MIT, training only), pydub (MIT), ffmpeg-python (Apache-2.0), FFmpeg (LGPL/GPL), FastAPI (MIT) and uvicorn (BSD-3-Clause) for the local API, pytest and ruff for tests and lint, uv for environments, Docker with Colima for the image, vast.ai A100s and Cloudflare R2 for the M5 runs. This site: a standard-library Python generator, Mermaid (MIT) rendered once to SVG, and no runtime dependency. [Source: the README](#).

THE TEAM

Created by Nathan, Hrushi & Tharakesh. Team Cross Exam at HackGT 13. Nathan (@nrstough) owned the deep detectors, the fold file, fusion, the orchestrator and the TSV; teammates owned the engineered detectors and the Docker image; Hrushi built the demo frontend against the pipeline's JSON contract. The component-by-component table of who decided what and what the AI wrote is in [CLAUDE.md](#).

Created by Nathan, Hrushi & Tharakesh.

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